Lecture plan · for the lecturer

Ethics, safe labs and CTF assessment

Cyber security · University · Lecture 4

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Teach the law and ethics.
- Build an isolated range.
- Assess CTFs fairly.

Activity: Build the safe lab

1. Warm-up (5 min): Read three short scenarios and ask students to vote: authorised or not?
2. Discuss (15 min): Work through the scenarios with the ACM Code and local law.
3. Draft (15 min): Teams draft a one-page rules-of-engagement sheet for a lab.
4. Design (20 min): Plan per-student flags, a write-up template and a rubric.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which scenario divided the room?
- What makes a lab truly isolated?
- How will you weight the fix?

Exit ticket

- Write the first line of your rules of engagement.

Try it in your next lecture
Before your first offensive lab, have every student sign a rules-of-engagement sheet and run the lab only on an isolated range.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Flags from your exercise appeared online. Which change protects the assessment?**

◯ Hide the scoreboard until the deadline ◯ New flags for next year's run

◯ Per-student flags and graded write-ups

2 **Design the lab network.**

Lab VMs use a _____ network with no route to the internet, and _____ let you reset them after each session.

Word bank: bridged · snapshots · host-only · passwords

3 **What belongs in a rules-of-engagement sheet? Choose all that apply.**

☐ What to do if a real flaw is found ☐ The exact permitted targets

☐ The legal consequences of unauthorised access

☐ Permission to publish any real flaw once the course ends

4 **If a student finds a real vulnerability in the university portal, the course should praise them for publishing it at once.**

◯ True ◯ False

5 **Put the safe offensive-lab sequence in order.**

☐ Sign the rules of engagement

☐ Teach the law and ethics

☐ Submit a graded write-up with the fix

☐ Attack the isolated range

Answers · for the lecturer

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1 **Per-student flags and graded write-ups**

Unique flags and write-ups make shared answers useless.

2 **Lab VMs use a host-only network with no route to the internet, and snapshots let you reset them after each session.**

Bridged networking would expose targets to the campus network.

3 **The exact permitted targets; What to do if a real flaw is found; The legal consequences of unauthorised access**

Scope, disclosure and law make expectations explicit.

4 **False**

Responsible disclosure reports privately first so the flaw can be fixed.

5 **Top to bottom: 2, 1, 4, 3**

Norms and consent before practice; reflection and defence after.

For you to think about

What would happen today if one of your students scanned a network outside your lab? What would you change?